

Provenance: VAST Challenge 2018
Title: **Mini-Challenge 1**

Overview

Mistford is a mid-size city located to the southwest of the Boonsong Lekagul Wildlife Preserve. The city has a small industrial area with four light-manufacturing endeavors. Mistford and the wildlife preserve are struggling with the possible endangerment of the Rose-Crested Blue Pipit, a locally loved bird. The bird's nesting pairs seem to have decreased alarmingly, prompting an investigation last year implicating Kasios Office Furniture, a Mistford manufacturing firm. Since the initial investigation, the situation has evolved: Kasios insists that they have done nothing wrong! They assert that grad student Mitch Vogel and his professors are mere media-seekers trying to draw attention away from their lackadaisical research. Kasios presents itself as an extremely eco-friendly organization. They have launched their own very public investigation into the issues raised last year and are reporting very different results! It's time to apply your visual analytics expertise to help illuminate the path to good science.

Mini-Challenge 1, "Cheep Shots?"

In 2017, the VAST Challenge results suggested that the Kasios Furniture manufacturing company (one of the firms located in Mistford) may have been a primary contributor to the apparent reduction of the number of nesting pairs of the Rose-Crested Blue Pipit, a favorite bird of Mistford residents and Boonsong Lekagul Nature Preserve visitors. Kasios supposedly used the banned substance Methylosmolene in their manufacturing process. They surreptitiously dumped process waste in the northeast region of the Preserve (mini-challenge 1 from 2017) and Methylosmolene was detected in their smokestack emissions (mini-challenge 2 from 2017).

Kasios now claims that the analysis was flawed and biased. To combat these conclusions, Kasios has launched their own "investigation" into the Pipit situation, and they are now reporting that there are plenty of Rose-crested Blue Pipits happily living and nesting in the Preserve. To back up this claim, they have provided a set of Pipit bird calls, recently recorded across the Preserve, with locations of where they were recorded. Clearly, they claim, the Pipits are a thriving population, and Kasios will provide even more supporting evidence as their investigation proceeds.

In last year's Challenge, an ornithology grad student from Mistford College named Mitch Vogel discovered the plight of the Pipit and carried out an investigation. Normally, we would call on Mitch again to help validate Kasios' claim. Unfortunately, Mitch is working far from Mistford in a remote area without internet access for an extended time and cannot be easily reached!

The Pangera Ornithology Conservation Society, who sponsored Mitch last year, is at their wit's end at what to do about this turn of events. The townsfolk and Preserve rangers seem satisfied that the recordings back up Kasios' claims. Mistford College does not have another Pipit expert they can call upon for help. But, they do have a collection of bird calls from the Preserve that has been vetted by various ornithology groups as having accurate identifications. They have heard that new techniques from machine learning and visual analytics can be applied to situations like this. Perhaps, the calls could be classified and analyzed using these technologies, and reviewed when Mitch returns.

Ground Truth

Of primary interest is that the bird calls provided by Kasios are not all Pipits. There are some Pipit calls, but there are also bird calls from other birds known to be in the region and also some not found in the Preserve at all (at least from the data provided to the contestants). The geolocation analysis of the various birds in the Preserve show some mixed results. It does appear that the Pipits move away from the contamination area from last year, but it would be difficult to argue for a catastrophic reduction in Pipit numbers just from detected calls from this data. Other species also move locations over time, but researchers need to take care in identifying trends as definitive or not.

Contestants that do report trends in movements or in bird populations should be careful not to claim anything that might be the result of sampling practices or yearly variations.

Data

Four datasets were provided to the contestants. The first is a dataset containing the birdcalls collected by Mistford College. There is also metadata about the bird calls as to their type and quality. There is a simple bitmap of the Preserve that can be used when considering geolocations, and finally, there are the calls claimed by Kasios to be all Pipits from across the Preserve.

Download the data using your Rowan account at

https://drive.google.com/file/d/1Obe5AQ7s6Yd5YNfSsOG14Rmq_WxefRA/view?usp=sharing

Contestant Questions

1. Using the bird call collection and the included map of the Wildlife Preserve, characterize the patterns of all of the bird species in the Preserve over the time of the collection. Please assume we have a reasonable distribution of sensors and human collectors providing the recordings, so that the patterns are reasonably representative of the bird locations across the area. Do you detect any trends or anomalies in the patterns? Please limit your answer to 10 images and 1000 words.
2. Turn your attention to the set of bird calls supplied by Kasios. Does this set support the claim of Pipits being found across the Preserve? A machine learning approach using the bird call library may help your investigation. What is the role of visualization in your analysis of the Kasios bird calls? Please limit your answer to 10 images and 1000 words.
3. Formulate a hypotheses concerning the state of the Rose Crested Blue Pipit. What are your primary pieces of evidence to support your assertion? What next steps should be taken in the investigation to either support or refute the Kasios claim that the Pipits are actually thriving across the Boonsong Lekagul Wildlife Preserve? Please limit your answer to 500 words.